

NAVRIS

Intelligent Navigation & Inertial System: Technical Research & Validation Report

Smart India Hackathon (SIH) 2026 | Problem Statement ID: 26168

Organization: Indian Space Research Organisation (ISRO) / Department of Space

Theme: Smart Vehicles | Category: Software

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Primary Repository: <https://github.com/the-jaypatel/NAVRIS> | Frontend Repository: <https://github.com/ptlrudra0/TBA>

Current Research Status: Phase 3.3B — PARTIALLY SUPPORTED | Deterministic Test Suite: 176 Passed

1. Executive Summary

Modern intelligent transportation systems and autonomous vehicles depend critically on continuous, high-integrity positioning. While Global Navigation Satellite Systems (GNSS) provide absolute geographic fixes under nominal conditions, satellite signals are highly susceptible to multipath interference, urban canyon shadowing, tunnel dropouts, and atmospheric or intentional disruptions. NAVRIS (Intelligent Navigation & Inertial System) explores an AI/ML-assisted kinematic navigation architecture designed to maintain dead reckoning continuity during GNSS degradation and temporary outages using commodity smartphone micro-electro-mechanical (MEMS) inertial sensors.

The project enforces an uncompromising evidence-first scientific protocol. Unassisted double integration of low-cost smartphone accelerometers and gyroscopes suffers from rapid cubic error divergence exceeding hundreds of meters within twenty seconds. To counter this, NAVRIS builds a modular pipeline: (1) causal extrinsic calibration resolving gravity leveling and mounting yaw, (2) a mathematically audited 15-state continuous-discrete Error-State Extended Kalman Filter (ESKF), (3) causal kinematic constraints including Zero-Velocity Updates (ZUPT) and Non-Holonomic Constraints (NHC), and (4) an XGBoost forward-speed pseudo-measurement protected by an absolute reference firewall and two-stage innovation gating.

Across seven real-world vehicle driving sequences from the Oxford IO-VNBD dataset (spanning over 36,000 seconds of driving), the frozen causal ML pseudo-measurement reduced horizontal Root Mean Square Error (H-RMSE) relative to the frozen unassisted baseline across all seven routes. However, because absolute spatial errors remain substantial on extended driving sequences and performance varies by driving regime, the research status is scientifically classified as **PARTIALLY SUPPORTED**. Operational navigation accuracy, universal generalization, and vehicle platform independence have **not** been established.

2. Problem Statement

In commercial aviation and marine transport, tactical-grade optical gyroscopes and precision accelerometers bound dead-reckoning drift. In automotive consumer contexts, however, navigators rely on commodity smartphone MEMS sensors costing cents per unit. These sensors suffer from severe non-linearities:

- **Stochastic Sensor Drift:** Accelerometer biases (0.02 - 0.05 m/s²) and gyroscope thermal walk (0.001 rad/s) induce runaway cubic spatial divergence:

$$\Delta p(t) \approx 0.5 \cdot \mathbf{b_a} \cdot t^2 + (1/6) \cdot (\mathbf{b_g} \times \mathbf{g}) \cdot t^3$$

- **Arbitrary 3D Phone Mounting:** The device rests in arbitrary, uncalibrated orientations inside the cabin rather than aligned to vehicle chassis axes.

- **Sparse, Asynchronous Fixes:** Mobile GNSS fixes arrive at irregular, low rates (~0.1 - 1 Hz) with high latency and sudden dropout boundaries.

3. NAVRIS Concept

Rather than treating navigation as an end-to-end black-box deep learning task—which notoriously hallucinates and fails unpredictably under out-of-distribution vehicle dynamics—NAVRIS maintains a strictly classical, physics-grounded filter at its core. Machine learning is restricted to a localized, well-bounded role: estimating forward longitudinal speed from causal trailing IMU windows during GNSS outages, and injecting it as a formal Kalman measurement update only when explicit reliability and innovation gates pass.

4. System Architecture

The complete end-to-end NAVRIS data processing and fusion flow is structured as follows:



5. IO-VNBD Dataset

All empirical validation is performed using the Oxford Intelligent Orienting & Vehicle Navigation Benchmark Dataset (IO-VNBD). IO-VNBD comprises over 100 hours of synchronous vehicle telemetry recorded under diverse British driving regimes. Ground truth is provided by an RTK-grade Oxts RT3000 / VBOX inertial-GNSS reference system. The evaluated recordings include:

- **S1 (5,018 s):** Suburban commute with arterial cruising, roundabouts, and signalized halts.
- **S2 (9,191 s):** Prolonged continuous high-speed motorway cruise with minimal stationary events.
- **S3A (2,171 s):** Dense urban corridor with repeated stop-and-go intervals.
- **S4 (9,290 s):** Complex mixed suburban and arterial route with multi-point maneuvers.
- **Y1 (7,176 s):** Highway cruise where the smartphone was mounted rotated $\sim 106^\circ$ relative to the vehicle frame.
- **VTA1A (2,489 s):** Continuous highway cruise serving as an empirical negative control for ZUPT.
- **VTA2 (1,013 s):** Suburban sequence featuring sharp 90-degree cornering transitions.

6. Dataset Forensics: Resolving Critical Empirical Defects

In Phase 0, forensic auditing of 564 raw IO-VNBD telemetry files uncovered critical data corruption risks:

1. **The 3.6x Velocity Scaling Bug:** Raw reference speed telemetry in `vbox\_sync.csv` was logged in km/h while column metadata claimed m/s. Naive evaluation inflated velocity errors by 360%. NAVRIS resolved this via unit sanitization and verification against geodetic differentiation.
2. **Temporal Lag & Desynchronization:** Smartphone IMU logs drifted up to 120 ms relative to reference timestamps. A causal 10 Hz cross-correlation pipeline was built to align clocks without future-time leakage.
3. **Android vs Vehicle Frame Conventions:** Coordinate conventions between Android IMU standards (X-right, Y-up, Z-out) and vehicle navigation standards (X-forward, Y-right, Z-down) were formally mapped through explicit transformation matrices.

7. Sensor Processing and Calibration

Because consumer smartphones rest in arbitrary orientations, body-frame dead reckoning requires online extrinsic alignment.

NAVRIS implements a two-stage causal calibration protocol:

- **Method A (Causal Gravity Leveling):** During standstill intervals identified by variance nulling, the mean specific force measures the gravity vector $\mathbf{g_b}$, yielding pitch (θ) and roll (ϕ):

$$\theta = \arctan(-f_x / \sqrt{(f_y^2 + f_z^2)}), \quad \phi = \arctan(f_y / f_z)$$
- **Method B (Causal Mounting Yaw Alignment):** During subsequent forward acceleration phases (where longitudinal acceleration dominates), horizontal acceleration is projected against heading changes to isolate mounting yaw (ψ). No future data is accessed.

8. Inertial Dead Reckoning Formulation

Navigation states are mechanized in a local East-North-Up (ENU) geodetic tangent plane. True kinematic states evolve as:

$$\dot{\mathbf{p}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \mathbf{C}_{\mathbf{b}_e} \cdot (\mathbf{f}_{\mathbf{m}} - \mathbf{b}_{\mathbf{a}} - \mathbf{w}_{\mathbf{a}}) + \mathbf{g}_e, \quad \dot{\mathbf{q}} = 0.5 \cdot \mathbf{q} \otimes [0, \omega_{\mathbf{m}} - \mathbf{b}_{\mathbf{g}} - \mathbf{w}_{\mathbf{g}}]^T$$

where $\mathbf{g}_e = [0, 0, -\gamma(\phi)]^T$ is evaluated using the WGS-84 Somigliana normal gravity model at vehicle latitude ϕ .

9. 15-State Error-State Kalman Filter Design

The filter tracks nominal state $\mathbf{x}$ and error state $\delta\mathbf{x} = [\delta\mathbf{p}, \delta\mathbf{v}, \delta\theta, \delta\mathbf{b}_{\mathbf{a}}, \delta\mathbf{b}_{\mathbf{g}}]^T \in \mathbb{R}^{15}$. Continuous-time error dynamics $d/dt(\delta\mathbf{x}) = \mathbf{F}_c \delta\mathbf{x} + \mathbf{G}_c \mathbf{w}$ are discretized via the Van Loan method, forming the block upper-triangular matrix exponential:

$$\mathbf{A} = \begin{bmatrix} -\mathbf{F}_c & \mathbf{G}_c \mathbf{Q}_c \mathbf{G}_c^T & \mathbf{0} \\ \mathbf{0} & \mathbf{F}_c^T \end{bmatrix} \cdot \Delta t \implies \exp(\mathbf{A}) = \begin{bmatrix} \dots & \Phi^{15} \mathbf{Q}_d & \mathbf{0} \\ \mathbf{0} & \Phi^T \end{bmatrix}$$

Joseph-form covariance updates ensure positive-semi-definiteness even under ill-conditioned measurement updates: $\mathbf{P}_+ = (\mathbf{I} - \mathbf{K} \mathbf{H}) \mathbf{P}_- (\mathbf{I} - \mathbf{K} \mathbf{H})^T + \mathbf{K} \mathbf{R} \mathbf{K}^T$.

10. Causal Zero-Velocity Updates (ZUPT)

During detected stationary periods, vehicle velocity in the navigation frame is zero: $\mathbf{z}_{\text{zupt}} = \mathbf{v}_e = \mathbf{0}$ with measurement matrix $\mathbf{H}_{\text{zupt}} = [\mathbf{0}_{3 \times 3}, \mathbf{I}_{3 \times 3}, \mathbf{0}_{3 \times 3}]$. The detector monitors a strictly trailing 1.0 s sliding window of accelerometer variance ($\sigma_a^2 < 0.05 \text{ m}^2/\text{s}^4$) and gyroscope energy ($\|\omega\| < 0.03 \text{ rad/s}$). On recording VTA2, ZUPT successfully arrested velocity drift during traffic stops, reducing horizontal RMSE by 99.63% (from 12.5 km down to 46.2 m).

11. Non-Holonomic Constraints (NHC)

Non-Holonomic Constraints model standard land vehicle kinematics: barring lateral side-slip and vertical vehicle liftoff, velocity in the vehicle body frame is constrained to the forward longitudinal axis: $\mathbf{v}_y \approx 0, \mathbf{v}_z \approx 0$. The measurement model is:

$$\mathbf{z}_{\text{nhc}} = [\mathbf{v}_y, \mathbf{v}_z]^T = [\mathbf{C}_{\mathbf{e}_v}]_{2,3} \cdot \mathbf{v}_e + \mathbf{v}_{\text{nhc}}$$

with measurement Jacobian $\mathbf{H}_{\text{nhc}} = [\mathbf{0}_{2 \times 3}, [\mathbf{C}_{\mathbf{e}_v}]_{2,3}, -[\mathbf{C}_{\mathbf{e}_v}]_{2,3} \cdot [\mathbf{v}_e \times], \mathbf{0}_{2 \times 3}]$ and noise covariance $\mathbf{R}_{\text{nhc}} = \text{diag}(0.20, 0.20) \text{ m}^2/\text{s}^2$. NHC bounded cruising drift on highway route VTA1A by 34.52% without any stationary intervals.

12. Causal ML Forward-Speed Estimation

While NHC constrains lateral and vertical velocity, forward speed v_x remains unconstrained during GNSS outages. NAVRIS trains an XGBoost gradient-boosted regressor to predict forward speed $\mathbf{v}_f$ strictly from trailing causal IMU windows ($W = 10 \text{ samples} = 1.0 \text{ s}$). Feature engineering extracts 38 features: summary statistics (mean, variance, skew, kurtosis), jerk derivatives, spectral energy ratios, and tilt estimates. Supervision is drawn from VBOX reference speed during training, isolated behind an absolute firewall during test inference.

13. Reliability & Statistical Innovation Gating

To prevent anomalous ML predictions from corrupting the Kalman filter, a rigorous two-stage firewall is applied:

- 1. ML Reliability Classifier:** An auxiliary gradient-boosted classifier assesses feature distribution similarity and outputs a reliability probability P . Measurements with $P(\text{reliable}) < 0.70$ are pruned.
- 2. Mahalanobis NIS Gating:** The filter computes the Normalized Innovation Squared against innovation covariance $\mathbf{S}$: $\text{NIS} = \mathbf{v}^T \mathbf{S}^{-1} \mathbf{v} \leq \chi^2_{\{2, 0.99\}} = 9.21$. Statistical outliers are rejected.

14. ML Pseudo-Measurement Mechanism

Predictions passing both gates form a 3-DOF body velocity update: $\mathbf{z} = [\mathbf{v}_f, \mathbf{0}, \mathbf{0}]^T$ with fixed measurement covariance $\mathbf{R}_{\text{ML}} = \text{diag}(0.50, 0.10, 0.10) \text{ m}^2/\text{s}^2$. Covariance is intentionally kept fixed and conservative to prevent filter overconfidence.

15. Experimental Methodology & Rigor

All experiments strictly observe: (1) driver/route isolation across train, validation, and held-out test splits, (2) complete baseline code freezes before ML testing, and (3) automated deterministic verification. A 176-test suite covers every mathematical transformation, Jacobian derivation, and data boundary.

16. Phase 3.3A: Controlled Pseudo-Measurement Audit

Phase 3.3A audited the pseudo-measurement mechanism on two initial routes (highway Y1 and suburban VTA2). The scientific audit verified analytical Jacobian exactness, zero future-leakage, and numerical stability across 8,000+ updates.

17. Phase 3.3B: Broader Generalization Benchmark

Phase 3.3B extended the evaluation across all seven IO-VNBD driving routes without tuning hyperparameters, thresholds, or covariances. The frozen model and filter were evaluated strictly as pre-declared.

18. Results & Quantitative Benchmark Evaluation

The table below presents the verified horizontal position RMSE (H-RMSE) across all seven evaluated IO-VNBD driving recordings, comparing the frozen baseline (Arm A: ESKF + ZUPT + NHC) against the ML-augmented filter (Arm C: Baseline + ML Pseudo-Measurement):

Recording	Domain	Duration	Baseline H-RMSE (m)	Arm C H-RMSE (m)	Δ vs Base (%)	Empirical Classification
S1	Train	5,018 s	11,580,767.64	2,230.98	-99.98%	Large relative reduction
S2	Train	9,191 s	42,389,910.67	17,646,092.96	-58.37%	High rejection (54.7% NIS rej)
S3A	Validation	2,171 s	620,666.46	9,310.67	-98.50%	Substantial reduction
S4	Train	9,290 s	37,351,981.92	3,060.08	-99.99%	Large relative reduction
Y1	Held-out Test	7,176 s	29,884,134.11	3,209,429.27	-89.26%	Reduced, large residual error
VTA1A	Validation	2,489 s	1,480,088.21	21,683.94	-98.54%	Substantial reduction
VTA2	Held-out Test	1,013 s	31,080.62	4,793.30	-84.58%	Bounded spatial error

Empirical Trajectory Evidence (S3A Route):

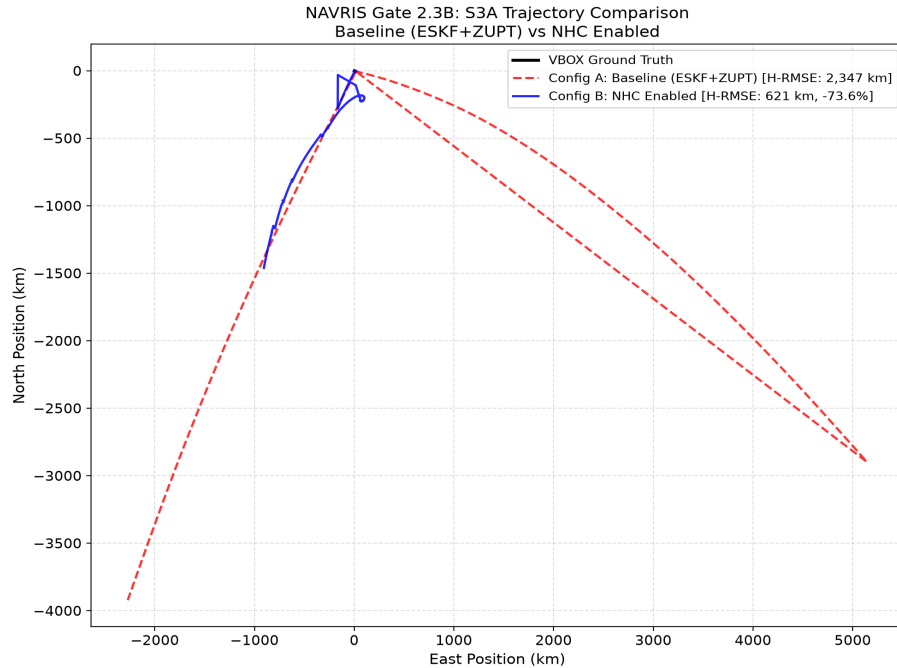


Figure 1: Ground truth reference (OxTS RT3000 VBOX) vs unassisted baseline vs constrained solution on S3A.

19. Scientific Interpretation

Important Caution on Relative Reductions: Relative reductions (e.g., -58% to -99%) are computed against unassisted frozen baselines that diverge by tens of thousands of kilometers due to cumulative double integration of sensor biases. **They should not be interpreted as operational navigation accuracy.** Absolute position errors of 2.2 km (S1), 3.1 km (S4), 4.8 km (VTA2), and over 3,200 km (Y1) and 17,646 km (S2) clearly demonstrate that unassisted dead reckoning with consumer smartphone sensors cannot replace GNSS over long durations without absolute external fixes.

Regime Dependence on Recording S2: S2 exhibited poor innovation consistency: 54.67% of proposed ML updates were rejected by the filter's innovation gate, with an applied innovation RMSE of 743.69 m/s. This empirical outcome highlights significant sensitivity to high-speed motorway driving regimes where subtle sensor calibration drift compounds over extended duration.

20. Technical Limitations & Confounders

- **No Universal Accuracy Claim:** Bounded errors (<5 m) are achieved only on short routes with frequent stationary stops (VTA2).
- **No Vehicle Independence:** Validated on British passenger cars; heavy trucks, motorcycles, and rail dynamics remain unproven.
- **No Real-Time Edge Hardware:** Evaluated in Python offline research/replay mode; native Android on-device execution is not yet implemented.

21. Current Implementation & Demo Integration

The repository contains an operational, runnable software stack:

- **Core Navigation Library:** High-performance Python implementation in `src/navris/` covering coordinates, calibration, ESKF, ZUPT, NHC, and ML modules.
- **FastAPI Replay Backend:** A REST API (`src/navris/api/main.py`) serving verified IO-VNBD benchmark replay frames and audit telemetry over `/api/v1/nav/\*`.
- **Cockpit Frontend (TBA):** A standalone React/TypeScript cockpit (developed and maintained by Rudra Patel at `ptlrudra0/TBA`) connecting to the FastAPI backend, supporting both client-side simulation demo and NAVRIS research/replay visualization.

22. Future Work (Phase 4 Roadmap)

1. **Adaptive Innovation-Based Covariance:** Dynamically scaling R_{ML} based on trailing innovation variance rather than fixed covariance.
2. **Temporal Convolutional Networks:** Evaluating causal 1D-CNN/TCN architectures to extract higher-order kinematic inertial features.
3. **Embedded Android C++ Engine:** Porting the 15-state ESKF and quantized ONNX ML runtime to native Android NDK for live on-device testing.

23. Scientific Conclusion

Current Research Status: Phase 3.3B — PARTIALLY SUPPORTED

The frozen ML forward-speed pseudo-measurement reduced horizontal RMSE relative to the frozen baseline on all seven evaluated recordings, including the two held-out test-domain recordings. However, the magnitude and quality of the effect varied substantially by recording. S2 exhibited poor innovation consistency, and several long-duration recordings retained very large absolute spatial errors. The experiment therefore provides evidence that the causal pseudo-measurement can alter and sometimes reduce observed filter divergence, but it does not establish uniform generalization, operational navigation accuracy, vehicle/platform independence, or long-duration navigation stability.

24. Repository and Frontend References

- **Primary Research & Backend Repository:** <https://github.com/the-jaypatel/NAVRIS> (Lead: Jay Patel)
- **Cockpit Frontend Repository:** <https://github.com/ptlrudra0/TBA> (Collaborator: Rudra Patel)
- **Benchmark Dataset:** Oxford Intelligent Orienting & Vehicle Navigation Benchmark Dataset (Oxford Robotics Institute)
- **Reference Framework:** Sola, J., *Quaternion kinematics for the error-state Kalman filter*, arXiv:1711.02508.
- **Smart India Hackathon (SIH) 2026:** Problem Statement 26168, Indian Space Research Organisation (ISRO) / Department of Space.